

Saqib Rashid Bhat

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building **AI agents, RAG systems, and ML applications** using **Python, LangGraph, FastAPI, and Scikit-Learn**. Built production-style solutions with **0.84 ROC-AUC, 5,000+ records, and 62/62 automated tests**, backed by **Docker and MLflow**.

TECHNICAL SKILLS

Languages: Python, SQL, Java

LLM / Generative AI: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Agentic AI, LangGraph, Prompt Engineering, Semantic Search, Sentence Transformers, Dense Embeddings, Vector Search, Information Retrieval, Tool Calling, LLM APIs, OpenAI-Compatible APIs

Machine Learning: Scikit-Learn, XGBoost, Random Forest, Logistic Regression, Classification, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, ROC-AUC

Backend Development: FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic

Databases: PostgreSQL, MySQL, FAISS (Vector Index)

Data Analysis: Pandas, NumPy, Data Preprocessing, Data Wrangling, Exploratory Data Analysis (EDA)

Tools & MLOps: Git, GitHub, Docker, Docker Compose, MLflow, Pytest, Postman

PROJECTS

Enterprise AI Operations Agent

GitHub

Python, LangGraph, FastAPI, SQLAlchemy, RAG, FAISS, Sentence Transformers, Pandas, Docker, Streamlit

- Built a production style **AI operations agent** using **LangGraph** to plan queries and route requests across **SQL, RAG, calculator, and data-analysis tools**, with evidence collection and response verification.
- Implemented an end-to-end **RAG pipeline using Sentence Transformers and FAISS** for enterprise policy retrieval, alongside SQL analytics over **5,000+ sales records**.
- Developed **evidence-grounded safeguards and refusal handling**; containerized the application with **Docker** and achieved **62/62 automated tests passing** across core components.
- Deployed the **FastAPI + Streamlit application on Render** for production-style demonstration.

Intelligent Document Q&A Assistant (RAG)

GitHub

Python, FastAPI, RAG, FAISS, Sentence Transformers, Groq (Llama-3.3-70B), PostgreSQL, MLflow, Streamlit, Docker

- Developed a **RAG-based Document Q&A Assistant** that ingests PDF, DOCX, and TXT files and answers questions using retrieved document context with source citations.
- Built the retrieval pipeline using **Sentence Transformers (all-MiniLM-L6-v2)** and **FAISS** to perform semantic search over document embeddings.
- Connected **Llama-3.3-70B (Groq)** through an OpenAI-compatible API to generate responses grounded in the retrieved content.
- Used **MLflow** to track experiments and monitor retrieval and response latency while improving overall pipeline performance.
- Exposed the application through a **FastAPI** backend and **Streamlit** interface, deploying it with **Docker Compose** and **PostgreSQL** for persistent conversation storage.

Customer Churn Prediction System

GitHub

Python, Scikit-Learn, FastAPI, Docker, MLflow, Pandas, NumPy, Joblib, Pydantic

- Built a **Customer Churn Prediction** model using a **Random Forest classifier**, achieving **0.84 ROC-AUC** and **80% accuracy** on telecom customer data.
- Developed a reusable preprocessing pipeline with **ColumnTransformer, OneHotEncoder, and StandardScaler** to ensure consistent data transformations during training and inference.
- Deployed the model as a **FastAPI** REST API with **Pydantic** validation, **Docker** containerization, and **MLflow** for experiment tracking and model reproducibility.

EXPERIENCE

Herway Coffee Roasters

June 2025 – June 2026

Operations, Data Reporting

Bangalore, India

- Managed inventory, billing, vendor coordination, and operational reporting using **Google Sheets** and **Swipe**, supporting day-to-day business operations.
- Prepared sales and inventory reports to support **data-driven planning** and operational decision-making.

EDUCATION

Cambridge Institute of Technology

2020 – 2024

Bachelor of Engineering in Computer Science

Bangalore, Karnataka

Minor in Artificial Intelligence & Data Science (IIT Mandi × Masai School)

2025 – 2026

Machine Learning, Natural Language Processing, Deep Learning, Probability & Statistics, Data Analytics

Online